

Chidubem Onwuchuluba

Atlanta, GA | chidubemonwuchuluba1@gmail.com | (346) 504-0993 | chidubemo.com
linkedin.com/in/chidubem-onwuchuluba | github.com/dubemmmm

Summary

Backend and applied AI engineer, currently an MS Computer Science student at Georgia State University (Machine Learning and Software Engineering). I've built payments APIs at Paystack, run a production real estate platform on AWS, and shipped agentic AI systems using A2A, MCP and the OpenAI Agents SDK.

Education

Georgia State University, Master of Science (MS) in Computer Science Aug 2025 - May 2027

- **GPA:** 3.92/4.0 • **Concentration:** Machine Learning and Software Engineering • Thesis track
- **Coursework:** Machine Learning, Computational Intelligence (Neural Networks, Optimization), Secure and Private AI, Cybersecurity and Ethical Hacking

Covenant University, Bachelor of Science (BS) in Computer Science Aug 2019 - May 2023

- **GPA:** 3.87/4.0, Magna Cum Laude, Top 2% • Ota, Nigeria

Technical Skills

Languages: Python (primary), C#, Go, Java, C++, SQL, JavaScript/TypeScript

Backend & Frameworks: Django, FastAPI, Flask, Celery, REST APIs, async Python, Next.js, React

Data Stores: PostgreSQL, MySQL, MongoDB, Redis, Pinecone, ChromaDB, Airtable

Cloud & DevOps: AWS (EC2, S3, CloudFront, ELB, CloudWatch), Azure, GCP, Docker, Kubernetes, GitHub Actions, Jenkins, GitLab CI, blue-green deploys

ML & AI: PyTorch, scikit-learn, CNNs, ResNet, FaceNet, adversarial ML (FGSM, PGD, C&W), RAG, hybrid retrieval (BM25 + dense + reranking)

Agentic AI: OpenAI Agents SDK, A2A protocol, MCP, multi-agent orchestration, tool use, agent evaluation

AI-Augmented Dev: Claude Code, Codex CLI, Cursor, GitHub Spec Kit, spec-driven development, AI code review, agent skill authoring (custom skills for Claude Code and Codex)

Automation & Testing: n8n, UiPath, Blue Prism, Palantir Foundry (Ontology, pipelines) • pytest, mypy (strict), deterministic test suites

Experience

Graduate Assistant | Georgia State University Aug 2025 - Present

- Teach and support 110+ undergraduate students in Computer Organization Programming, running labs, debugging student code and configuring Linux/Windows lab environments with 95% student satisfaction
- Wrote Python automation scripts that streamlined grading and lab administration, freeing time for one-on-one student support
- Design and lead thesis and dissertation formatting workshops for graduate students across the College of Arts and Sciences, walking them through APA and ASA template requirements
- Translate technical requirements (document structure, styles, pagination, PDF output) into plain, step-by-step guidance for students from non-technical disciplines
- Built and shipped a thesis/dissertation format compliance checker that cut my review time by 70%+ at 94% accuracy
- Designed it as a deterministic two-tier checker (XML-structural checks plus PDF-rendered geometry) backed by a rules manifest of 48 verified evidence checks, with mandatory LibreOffice rendering so every finding is visually verified
- Wrapped the checker in a FastAPI web app with a student-facing upload interface so students can self-check before submission

Software Engineer | Paystack (YC W16, Stripe) May 2024 - Aug 2025

- Built and optimized RESTful APIs in a platform processing 2M+ daily transactions at 99.9% uptime; implemented a Redis caching layer that reduced response time by 45%
- Contributed to a multi-tenant SaaS architecture serving 50,000+ merchants, building containerized microservices on Docker and Kubernetes
- Improved CI/CD pipelines (Jenkins, GitLab CI), raising deployment frequency by 60% with zero-downtime blue-green releases
- Profiled slow queries across high-throughput payment flows and reduced p99 latency through index tuning and query restructuring
- Worked in C# alongside Python, across a mixed AWS and Azure footprint, collaborating with PMs and cross-functional teams from requirements through production

Software Engineering Intern | Rotawn Energy Jan 2024 - May 2024

- Built data visualization dashboards for requisition and expense tracking, enabling real-time reporting that contributed to a 3% cost reduction in 6 months
- Developed and deployed a Visitor Management web app with QR-based authentication and SMS notifications, cutting manual processing time by 40%
- Maintained 99.5% uptime through proactive monitoring (AWS CloudWatch), database query optimization and automated backup pipelines

Projects

CW Lagos Real Estate Operations Platform | *Django, PostgreSQL, Celery, AWS, Airtable, Pinecone, Next.js*

Website

- Architected a full-stack cloud platform on AWS (EC2, S3, CloudFront, ELB, CloudWatch) managing 300+ properties and 700+ monthly visitors, with auto-scaling, CDN and monitoring
- Optimized PostgreSQL queries and added connection pooling and load balancing, reaching 98% uptime and 2-second page loads while cutting infrastructure cost by 25%
- Diagnosed recurring OOM kills on a t3.micro through memory profiling and right-sized to t3.medium, ending production crashes
- Built a Celery-driven Airtable sync pipeline; fixed a crash on blank Progress Status fields, eliminated duplicate sync scheduling, and corrected pruning logic so stale records were removed safely
- Added Pinecone vector search for semantic property lookup (earlier version: RAG over listings with ChromaDB)
- Set up CI/CD with GitHub Actions and n8n integration workflows
- Designing a dark, instrument-grade real estate intelligence dashboard for the operations team

Multi-Agent Trip Planning Network | *Python (asyncio), FastAPI, A2A, MCP, OpenAI Agents SDK*

- Built specialist agents as independent FastAPI services that discover each other and delegate work over the Agent2Agent (A2A) protocol, to show agents from different vendors can interoperate
- Exposed external tools and data through MCP servers, keeping tool access separate from agent reasoning
- Designed the protocol layer to be framework-agnostic, so an agent built on a different SDK can join the network without changes elsewhere
- Wrote the system in async Python end to end so agent-to-agent and tool calls run concurrently
- Built a 351-test suite, including 287 deterministic tests that run without live LLM calls, and kept the codebase clean under mypy strict

Defending End-to-End Face Authentication Pipelines | *PyTorch, ResNet-18, FaceNet, DeePixBiS*

- Studied the identity gate used in digital KYC onboarding at banks, fintechs and payment platforms: a liveness (anti-spoofing) model chained with a face recognition model, which an attacker can fool in one shot by targeting both stages
- Built an end-to-end pipeline with a ResNet-18 face anti-spoofing model gating a FaceNet face recognition model
- Implemented pipeline-aware attacks against both stages: FGSM, PGD, Carlini-Wagner (C&W), and a joint multi-objective PGD that defeats liveness detection and identity matching at the same time
- Proposed an internal-activation fingerprinting defense: a 1,411-dimensional feature vector drawn from the pipeline's intermediate layers, used to flag adversarial inputs before an authentication decision is made
- Reached 0.997 AUROC detecting adversarial inputs across attack types
- Tested cross-paradigm transferability against DeePixBiS, a depth-based liveness model, to check whether attacks and the defense generalize beyond one architecture
- Extended earlier work on an adaptive multi-stage detector that escalates analysis only for harder inputs (85–93% detection at 6.4 average forward passes); co-authored an IEEE-format paper with Cherisha Killari Giribabu

AI-Powered Telegram Personal Assistant | *Python, OpenAI Agents SDK, Telegram Bot API, Google OAuth 2.0, APScheduler*

- Designed a dispatcher-plus-specialists architecture: a routing agent reads each message and hands off to one of 7 specialist agents (Gmail, Calendar, Tasks, web search, reminders, notes, macOS control)
- Built the Gmail agent with correct MIME threading so replies land in the right conversation
- Built Calendar and Tasks agents with full create, read, update and delete, using ISO 8601 handling for reliable dates and time zones
- Implemented Google OAuth 2.0 with automatic token refresh so the assistant stays authorized without repeated sign-ins
- Added live web search through the Brave Search API for questions that need current information
- Built stateful per-user conversation memory with 30-minute session expiry, so multi-turn requests keep their context
- Built async push reminders with APScheduler so the bot messages the user proactively at the scheduled time, plus a macOS control agent that runs local system actions from chat

Autonomous Multi-Agent Code Reviewer (CLI) | *Python, OpenAI Agents SDK, asyncio, Pydantic*

GitHub

- Built four reviewer agents with distinct lenses: StaticAnalyzer (quality, style, anti-patterns), SecurityAuditor (OWASP Top 10, injection, hardcoded secrets, CVE lookup), PerformanceReviewer (complexity, N+1 patterns, memory leaks, async misuse) and DocChecker (docstrings, type hints, naming, dead code)
- Ran all four concurrently with `asyncio.gather`, so a full four-angle review takes about as long as a single agent call
- Built a fifth ReportAggregator agent that deduplicates overlapping findings, ranks issues by severity, and produces a 0–100 score, verdict, executive summary and top 3 priority actions
- Enforced structured agent output with Pydantic schemas so every review is machine-readable and consistent
- Gave agents file-reading tools plus a Brave Search tool for live CVE and performance benchmark lookups
- Streams each agent's findings to the terminal as it finishes and saves a full Markdown report; supports Python, JS, TS, JSX/TSX, HTML, CSS, JSON and YAML

Claude Code, Codex

- Built a browser-based assessment platform that scores how engineers investigate, prompt, verify and decide when to trust AI agents, with assessor, candidate and reviewer flows
- Provisioned an isolated E2B sandbox VM per session in about 5 seconds: clones the repo, installs dependencies, pre-installs and pre-authenticates Claude Code and Codex, and allocates a real PTY
- Built the candidate IDE: file tree, Monaco editor with autosave, xterm.js terminal on the sandbox PTY, and an in-app AI chat (OpenRouter) with a tool-calling agent loop streamed over NDJSON
- Instrumented every action into Supabase Postgres: keystroke edits, time per file, terminal commands and output, test runs, chat prompts with token counts, and Claude Code tool calls captured through hooks; secrets regex-redacted before persistence
- Designed a deterministic Agentic Coding Score with four weighted pillars (Outcome 40%, AI Effectiveness 25%, Comprehension 20%, Process 15%), where every signal traces to logged events and no LLM judges the candidate
- Measured codebase exploration, prompt quality (volume, substance, tool variety, edit acceptance rate) and process hygiene (iterative testing, verifying after edits, not weakening tests)
- On submit, auto-runs tests, snapshots the git diff and harvests Claude Code hook logs; the reviewer page shows the score, auto-generated red flags, per-pillar breakdown, diff and a filterable timeline

Intelligent Waste Classification System | *PyTorch, CNN, Django, PostgreSQL, Redis,*

GitHub

Docker, AWS

- Trained a custom CNN on 50,000+ images, reaching 92% accuracy across 6 material categories with real-time inference serving up to 100+ predictions
- Built an end-to-end pipeline for preprocessing, training and evaluation, with ablation studies on architecture variants
- Engineered a REST API with Redis caching and geospatial indexing that matches users to 20+ recycling facilities, cutting search time by 65%

Research and Writing

- C. Onwuchuluba, C. Killari Giribabu. “Defending End-to-End Face Authentication Pipelines via Internal-Activation Feature Fingerprinting.” IEEE-format paper, Georgia State University.
- C. Onwuchuluba, C. Killari Giribabu. “Adaptive Multi-Stage Defense Against Adversarial Examples: Progressive Intensity Matching Input Difficulty.” CSC 8230, Georgia State University, Spring 2026.
- MS thesis, Georgia State University (in progress)

Hackathons and Awards

HackATL 2025 Finalist: PulseGuard | *Machine learning, sensor data, real-time dashboards*

2025

- Built a predictive safety wearable in 48 hours that runs an ML risk model on sensor data to detect hazardous chemical exposure in real time
- Built an automated alerting dashboard that flags at-risk users before exposure becomes dangerous

Certifications

AWS Cloud Practitioner • GitHub Certified: Agentic AI Developer (Microsoft) • IBM AI Developer • Deep Learning with PyTorch (IBM)